

Networks Decide for Us

When homophily governs the criticality of social adoption

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The question

- Collective behaviors (innovations, mobilizations) spread as **adoption cascades** on social networks.
- Most models use *synthetic* topologies and treat agents as cognition-free relays [1].
- Real networks are shaped by **multidimensional homophily** — similar people cluster [2].

Conditional on *fixed individual characteristics*,

Can the **shape of realistic networks alone** change the outcome?

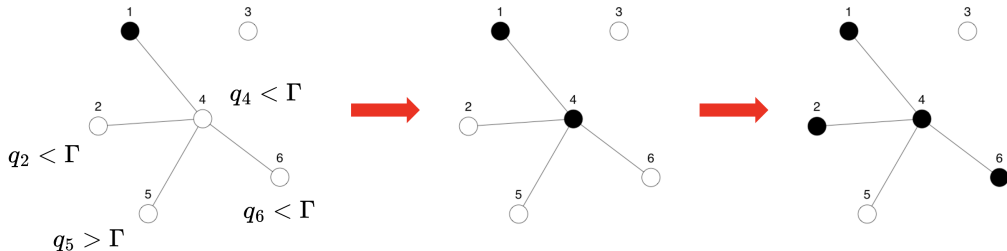
Can the structure itself be an **autonomous force**, deciding not just *how much* spreads, but *how* it spreads?

The model

The model — 1) Rational Choice

- The attractiveness of the innovation is its *Intrinsic Utility Level* (IUL), $\Gamma \in [0, 1]$.
- Each person has a *Minimum Utility Requirement* (MUR), $q_i \in [0, 1]$, representing their aversion to adopting the behavior.

An agent adopts on its own iff $q_i \leq \Gamma \implies a_i = 1$

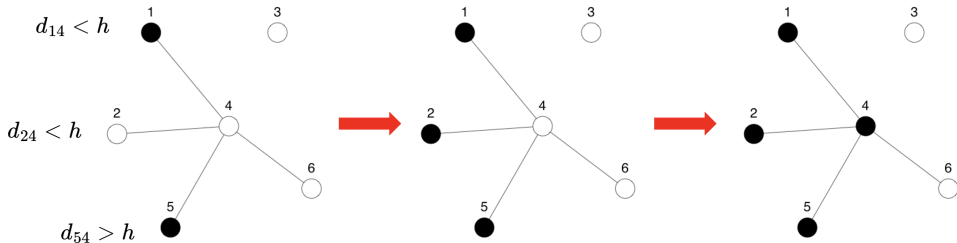


The model — 2) Selective Social Influence

Adopt by peer contagion if $\tau_i \leq \tilde{E}_i$ [5,6] — but only those socially *close* [7] count:

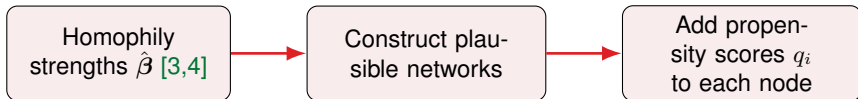
$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \quad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij})$$

- Where **MSP** (Maximum Social Proximity) sets how far influence reaches.



Imputing structure & running the model

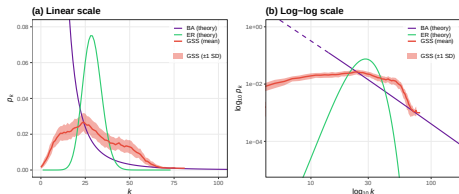
Imputing a network onto a survey



Tie probability is a function of social distance (age, sex, education, race, religion):

$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \beta^\top \mathbf{d}_{ij}$$

$\hat{\beta}$ comes from GSS confidant data [4]; propensities q_i come from real survey answers (innovation, collective action).



Right-skewed degree structure that Erdős-Rényi / Barabási-Albert models fail to match jointly.

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR` [9], over plausible **GSS** vs. degree-preserving randomized **GSS-DP** topologies.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

Total: $41 \times 13 \times 4 \times 4 \times 5 \times (\text{runs}) \approx 4 \times 10^6$ **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{DP}} \mathbb{I}_{\text{DP}} + s(\text{IUL}, \text{MSP})$$

A statistical-physics reading

Mapping social adoption to a ferromagnet

Ising system	NDFU
Spin $s_i \in \{0, 1\}$	$\leftrightarrow$ Adoption state a_i
Magnetization M	$\leftrightarrow$ Adoption $\Phi = N_{\text{ad}}/N$
External field H	$\leftrightarrow$ IUL
Temperature T	$\leftrightarrow$ MSP
Exchange J_{ij}	$\leftrightarrow$ Clustering w_{ij}
Susceptibility χ	$\leftrightarrow$ $\text{Var}(\Phi)$

Pseudo-Hamiltonian:

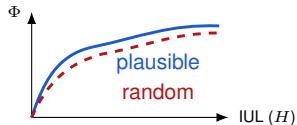
$$\mathcal{H} \approx - \sum_{\langle i,j \rangle} w_{ij}(h) s_i s_j - \Gamma \sum_i s_i$$

Low MSP $\rightarrow$ rigid echo chambers
(frozen domains).

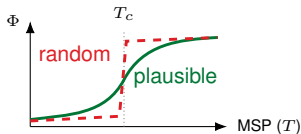
High MSP $\rightarrow$ boundaries “melt”, global
cascades possible.

Susceptibility diverges near criticality [8]: $\chi = \text{Var}(\Phi) \propto |MSP - MSP_c|^{-\gamma}$

Two paths through the critical point



Path 1 — vary H at fixed T : exponent δ .



Path 2 — vary T : a continuous (2nd-order) or discontinuous (1st-order) transition; exponent γ .

Path 1 — fix MSP, vary IUL (H)

- the field route (critical isotherm)
- yields the exponent δ

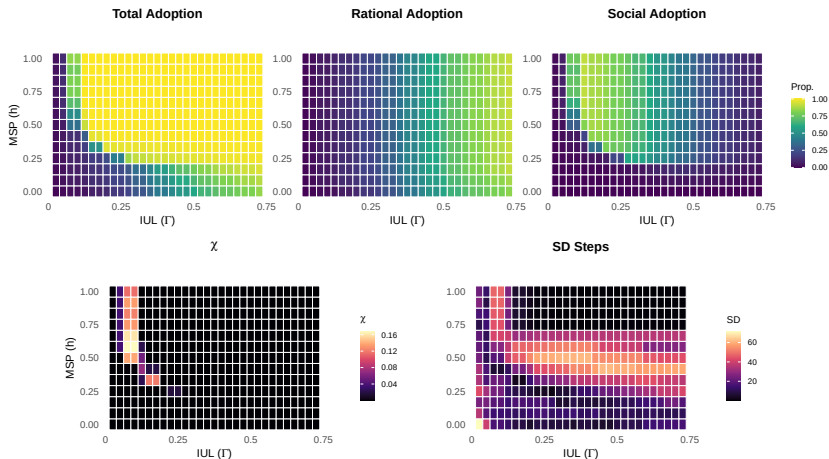
Path 2 — fix IUL, vary MSP (T)

- the temperature route to the critical point
- if continuous,
 $\chi = \text{Var}(\Phi) \sim |MSP - MSP_c|^{-\gamma}$

What does each path reveal about the structure? → **Results.**

Results

Result 1 — Adoption and criticality (dynamics)



Top: Total, Rational and Social adoption. Rational adoption grows smoothly with IUL.

Bottom-left: Susceptibility $\chi = \text{Var}(\Phi)$; **Bottom-right:** relaxation-time SD divergence.

Result 2 — A structural premium (volume)

BAM isolates the marginal effect of destroying empirical structure:

$$\beta_{DP} = -0.850^{***} \text{ (OR} = 0.43\text{)}$$

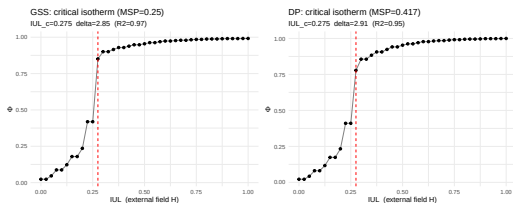
→ randomization cuts adoption by ~**57%**

Under *identical* propensities, homophilic clustering lets social reinforcement build inside **local echo chambers**.

Total adoption (GSS)	Estimate	OR
(Intercept)	6.534***	—
τ_{mean}	-6.944***	—
τ_{sd}	5.942***	—
<i>Seeding (ref: Random)</i>		
Central	0.120***	1.127
Closeness	0.119***	1.126
Eigenvector	0.118***	1.125
Marginal	-0.067***	0.935
<i>Topology (ref: Plausible)</i>		
Deg.-pres. random	-0.850***	0.427
Deviance explained: 82.2% $N = 4,093,440$		

*** $p < 0.001$. OR = $\exp(\beta)$.

Result 3 — Field direction: $\delta \approx 3$ for both networks

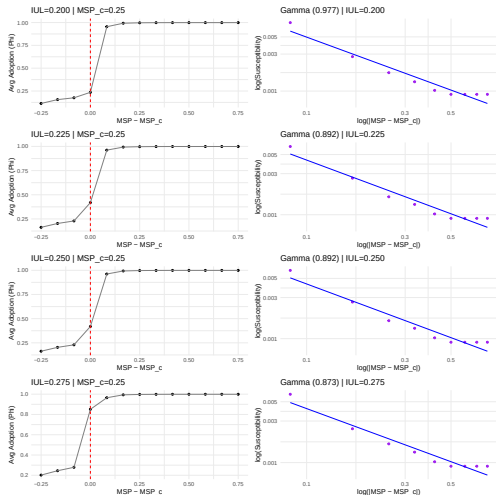


Critical isotherms $\Phi(IUL)$ at $MSP = MSP_c$.

Topology	δ
Plausible (GSS)	2.85
Randomized (GSS-DP)	2.91
mean-field	3.0

- Sweeping IUL at $MSP = MSP_c$, *both* networks give $\delta \approx 3$ — close to mean-field.
- The field (IUL) direction does **not** distinguish the two topologies.

Result 3 — Universality exclusive to realistic networks



IUL	Plausible γ	Randomized γ
0.200	0.98	3.71
0.225	0.89	3.67
0.250	0.89	3.67
0.275	0.87	4.67
mean	0.91	3.93

- Only the realistic networks reaches $\gamma \approx 1$ — continuous trans. — *Mean-Field universality*.
- Randomization $\Rightarrow \gamma \approx 4$. Rewire the network and the mean-field behavior is lost.

$$\chi \sim |MSP - MSP_c|^{-1}.$$

Conclusions

Conclusions

Can the shape of realistic networks *alone* change the outcome?

- **Volume:** empirical homophily yields a **structural premium** (+57% odds).
- When society is not too restrictive (high MSP), **highly attractive and unattractive innovations reach nearly the same adoption** — purely through social influence.

Is structure an autonomous force deciding *how* it spreads?

- **Dynamics:** on a realistic network the *aggregate* pattern is **continuous**.
- Aggregate phenomenon follows the familiar **Rogers S-curve**, across very different diffusions as fax machines, hashtags, and protest waves.

*Macroscopic social change is not the sum of individual preferences
— it's governed by the structure of our real topology.*

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Thanks!



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<https://doi.org/10.5281/zenodo.19360209>

Appendix — The mean-field paradox & its resolution

Puzzle

Mean-field $\gamma = 1$ usually needs a *well-mixed* system. But we have a *strongly clustered, homophilic* network.

Resolution

Complex networks have no finite spatial dimension — neighborhoods expand exponentially, so the system sits *above* the upper critical dimension and mean-field is the *generic* expectation. The real contrast is the *order* of the transition: homophilic clustering yields a continuous (2nd-order) transition, whereas degree-preserving randomization collapses into a discontinuous (1st-order) regime shift.

Homophily gives the system a “**sociological viscosity**”: across comparable societies, tipping points carry *statistical* early warning (rising variance, slow recovery) instead of detonating blind.